# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

**Department of Computer Science and Engineering**

## CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

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|---|---|---|---|
| **Course Code:** | CSA10 | **Course Title:** | Software Engineering |
| **CO Assessed:** | CO2 | **Assessment:** | Assessment Tool 2 – Scenario-Based Requirement Analysis |
| **Weightage:** | 20% | **Total Marks:** | 25 |
| **CO2 Statement:** | *Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills* | | |
| **Student Name:** | M.HASHETHA | **Reg. No.:** | 192521162 |
| **Date:** | 04-08-2026 | **Duration:** | 60 minutes |

**Code of Conduct**

*I __M.HASHETHA_(192521162)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _M.HASHETHA_

## Annexure A – Scenario-Based Requirement Analysis

**Instructions to Students:**

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

## Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
    - Carefully understand the given problem statement.
    - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
    - List all functional requirements required by the proposed system.
    - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
    - Identify all primary and secondary actors interacting with the system.
    - List the major use cases associated with each actor.
    - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**

    Develop an SRS document containing:
    - Introduction
    - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

### Rubrics for Calculation

| Criteria | Excellent (4 Marks) | Good (3 Marks) | Satisfactory (2 Marks) | Needs Improvement (1 Mark) |
|---|---|---|---|---|
| **Scenario Understanding and Problem Analysis** | Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis. | Demonstrates good understanding with minor omissions. | Basic understanding with limited analysis. | Poor understanding; major aspects missing or incorrect. |
| **Functional and Non-Functional Requirement Identification** | Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements. | Identifies most requirements with minor classification errors. | Identifies only basic requirements; several omissions. | Requirements are incomplete, inaccurate, or poorly classified. |
| **Actor Identification and Use Case Modeling** | Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards. | Most actors and use cases identified; diagram has minor errors. | Limited actors/use cases identified; diagram partially correct. | Actors/use cases missing or incorrect; diagram absent or inaccurate. |
| **Software Requirement Specification** | SRS is well-structured, complete, professionally organized, and | SRS is organized with minor missing sections or formatting issues. | SRS includes only essential sections with limited detail. | SRS is incomplete, poorly organized, or |

| | | | | |
|---|---|---|---|---|
| **(SRS) Documentation** | includes all required sections. | | | lacks essential components. |
| **Assumptions, Constraints, and Requirement Validation** | Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification. | Identifies most assumptions and constraints; validation is adequate. | Limited assumptions/constraints; validation is superficial. | Assumptions, constraints, and validation are missing or incorrect. |

| Criteria | **Mark** |
|---|---|
| Scenario Understanding and Problem Analysis | /5 |
| Functional and Non-Functional Requirement Identification | /5 |
| Actor Identification and Use Case Modeling | /5 |
| Software Requirement Specification (SRS) Documentation | /5 |
| Assumptions, Constraints, and Requirement Validation | /5 |
| **TOTAL** | /5 |

# SIMATS ENGINEERING

ASSESSMENT TOOL-2

CO-1

Assignment Title: **Scenario-Based Requirement Analysis**

AI-Powered Smart Court Case Management and Judicial Analytics Platform


COURSE: SOFTWARE ENGINEERING

COURSE CODE:CSA-1017

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# 1. Introduction

The **AI-Powered Smart Court Case Management and Judicial Analytics Platform** is an advanced software solution designed to modernize and improve the efficiency of judicial systems through digital transformation. Courts across the world face significant challenges due to increasing numbers of legal cases, manual documentation, inefficient hearing scheduling, and limited analytical support. These issues often result in delayed justice, increased case backlogs, poor workload management, and reduced transparency in judicial processes. Traditional court management systems mainly focus on storing case records and offer limited support for intelligent decision-making or automation, making it difficult for judicial authorities to handle growing workloads effectively.

This proposed system leverages **Artificial Intelligence (AI), Natural Language Processing (NLP), Cloud Computing, and Data Analytics** to automate and optimize various court operations. AI techniques are used to classify legal documents, predict case completion timelines, and analyze historical case data, while NLP helps in understanding and categorizing legal documents efficiently. Cloud technology provides secure and scalable storage for case records, enabling authorized users to access information anytime and from any location. Data analytics offers meaningful insights into judicial performance, case trends, and workload distribution, allowing administrators to make informed decisions.

The platform provides an integrated environment for judges, lawyers, court clerks, court administrators, citizens, and system administrators. Lawyers can register cases and upload legal documents online, court clerks can verify documents and schedule hearings, judges can monitor assigned cases and update judgments, while administrators can oversee judicial workload and generate analytical reports. Citizens can also track the status of public cases, improving transparency and accessibility within the judicial system.

The primary objective of the platform is to digitize court case management while reducing manual effort and improving operational efficiency. It aims to automate legal document classification, intelligently schedule hearings, predict case completion durations, analyze judicial workloads, generate comprehensive legal reports, and reduce case backlogs. By integrating modern technologies into judicial administration, the system promotes faster case processing, better resource utilization, enhanced transparency, improved public trust, and data-driven decision-making.

Overall, the AI-Powered Smart Court Case Management and Judicial Analytics Platform serves as a comprehensive LegalTech solution that enhances the quality of judicial services.

# 2. Scenario Understanding and Problem Analysis

## 2.1 Overview of the Scenario

The judicial system plays a vital role in maintaining law, justice, and public confidence. However, many courts continue to rely on manual processes and traditional methods for managing legal cases, resulting in delays, inefficiencies, and an increasing backlog of pending cases. As the number and complexity of cases continue to grow, it becomes challenging for judges, court staff, and legal professionals to efficiently manage case records, schedule hearings, and monitor judicial workloads. Existing court management systems primarily focus on storing case information and provide very limited support for intelligent automation, predictive analytics, and data-driven decision-making.


The proposed AI-Powered Smart Court Case Management and Judicial Analytics Platform aims to transform judicial administration by integrating Artificial Intelligence (AI), Natural Language Processing (NLP), Cloud Computing, and Data Analytics into a single digital platform. The system automates legal document classification, intelligently schedules hearings, predicts case completion timelines based on historical data, and provides comprehensive dashboards for monitoring judicial performance and workload. By replacing manual operations with automated workflows, the platform enhances efficiency, reduces processing time, and improves transparency throughout the judicial process.

## 2.2 Business Domain

The proposed system belongs to the Legal Technology (LegalTech) and Judicial Administration domain. LegalTech focuses on applying modern technologies to improve legal services and court operations. The platform supports digital transformation within the judiciary by enabling secure management of legal records, intelligent case handling, analytical reporting, and improved public access to justice. It serves as a centralized solution for courts, judges, lawyers, administrators, and citizens, helping judicial institutions operate more efficiently and transparently.

## 2.3 Stakeholders

The successful implementation of the system involves multiple stakeholders, each with specific roles and responsibilities.

Primary Stakeholders

- Judges – Review assigned cases, conduct hearings, deliver judgments, and monitor case progress.
- Lawyers – Register new cases, upload legal documents, monitor case status, and receive hearing notifications.


- Court Clerks – Verify submitted documents, maintain case records, and schedule hearings.
- Court Administrators – Allocate cases, monitor judicial workload, generate reports, and oversee court operations.
- Litigants (Citizens) – Track the status of their cases and receive updates regarding hearings and judgments.
- System Administrators – Manage user accounts, maintain the system, perform backups, and ensure data security.

Secondary Stakeholders

- Government Judicial Department
- AI and NLP Processing Engine
- Cloud Service Provider
- Notification Services (Email/SMS)

**2.4 Existing Challenges**

The current judicial process faces several operational and technological challenges that reduce overall efficiency.

- Manual registration and maintenance of case records.
- Paper-based documentation leading to document loss and duplication.
- Delays in scheduling hearings due to manual coordination.
- Increasing backlog of pending cases.
- Uneven distribution of workload among judges.
- Lack of intelligent document classification.
- Limited analytical tools for monitoring court performance.

- Difficulty in predicting case completion timelines.
- Lack of transparency for lawyers and citizens regarding case progress.
- Time-consuming report generation and administrative tasks.

## 2.5 Problem Statement Analysis

The major problem addressed by this project is the inefficiency of traditional court case management systems. Manual documentation, inefficient scheduling, and the absence of intelligent decision-support systems significantly slow down judicial processes. Courts struggle to prioritize cases, allocate judges effectively, and monitor overall performance. As a result, case backlogs continue to increase, delaying justice and reducing public confidence in the judicial system.


The proposed AI-powered platform addresses these challenges by automating repetitive tasks, intelligently managing legal documents, optimizing hearing schedules, predicting case completion durations, and providing real-time analytical insights for judicial administrators. This enables courts to deliver faster, more transparent, and more efficient legal services.

## 2.6 System Objectives

The primary objectives of the proposed system are:

- Digitize the complete court case management process.
- Automate legal document classification using AI and NLP.
- Predict estimated case completion timelines.
- Optimize hearing schedules to reduce delays.
- Analyze judicial workload for better resource allocation.
- Generate analytical reports for administrators.
- Improve transparency by allowing stakeholders to track case progress.
- Reduce the backlog of pending cases.
- Enhance decision-making through data analytics.
- Improve the overall quality and accessibility of judicial services.

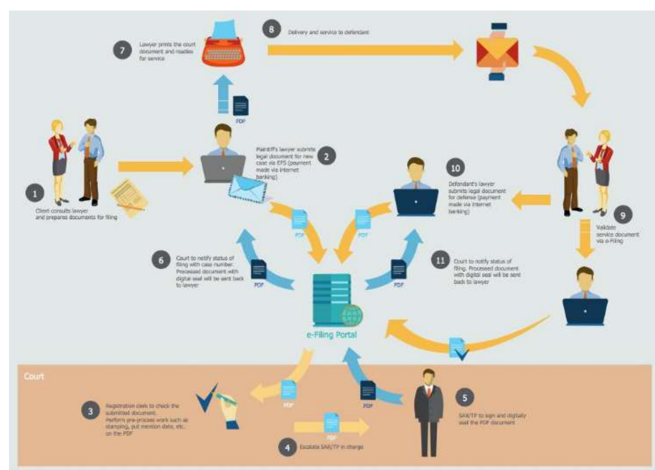
### 2.7 Expected Benefits

Implementing the proposed system provides significant benefits to both judicial authorities and the public.

- Faster processing of legal cases.
- Reduced case backlog.
- Efficient allocation of judges and court resources.
- Improved transparency and accountability.
- Better workload management.
- Accurate prediction of case completion times.
- Secure digital storage of legal documents.
- Improved public access to case information.
- Data-driven decision-making through analytical dashboards.
- Enhanced efficiency and quality of judicial administration.

## 3. Scope of the System

The **AI-Powered Smart Court Case Management and Judicial Analytics Platform** is designed to provide a comprehensive digital solution for managing court cases throughout their entire lifecycle, from case registration to final judgment and closure. The system aims to replace traditional manual processes with an intelligent, automated platform that enhances efficiency, transparency, and accessibility within the judicial system. It enables judges, lawyers, court clerks, administrators, and citizens to perform their respective tasks through a centralized web-based application.




The platform supports **online case registration**, allowing lawyers and authorized court staff to file new cases electronically and upload legal documents securely. Using **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)**, the system automatically classifies legal documents based on their content, reducing manual effort and improving document organization. Court clerks can verify submitted documents, while administrators can assign cases to judges based on workload, specialization, and availability.

A key feature of the system is its **intelligent hearing scheduling module**, which minimizes scheduling conflicts by considering judge availability, courtroom resources, and case priority. The platform also provides **AI-based prediction of case completion timelines** by analyzing historical case data and current case characteristics. This helps judges and court administrators estimate delays, prioritize urgent cases, and improve overall judicial planning.

The system includes **real-time case tracking**, enabling lawyers and litigants to monitor case status, hearing dates, judgments, and other important updates through a secure portal. Automated notifications via email or SMS ensure that all stakeholders receive timely information about hearings, document submissions, and case progress.

To support effective judicial administration, the platform provides **interactive dashboards and analytical reports**. These dashboards display information such as pending cases, disposed cases, judge workload, hearing schedules, and court performance metrics. Administrators can generate detailed reports to analyze trends, identify bottlenecks, and make informed decisions for improving court operations.

The scope of the system also includes **secure user authentication, role-based access control, cloud-based data storage, audit logging, and backup mechanisms** to ensure data integrity, privacy, and compliance with legal regulations. Citizens can access publicly available case information without compromising confidential legal records, thereby improving transparency and public trust in the judicial process.

However, the proposed system does **not** include the actual legal decision-making process, judgment drafting, or legal interpretation by AI. Judicial decisions remain entirely under the authority of judges. The platform serves only as a decision-support and case management system, assisting judicial personnel through automation, analytics, and efficient workflow management.

Overall, the AI-Powered Smart Court Case Management and Judicial Analytics Platform provides a scalable, secure, and intelligent solution that modernizes court operations, reduces case backlogs, improves resource utilization, and enhances the delivery of judicial services.

## 4. Functional Requirements

Functional requirements describe the specific functions and services that the **AI-Powered Smart Court Case Management and Judicial Analytics Platform** must perform to meet the needs of its users. These requirements define how the system should behave and what operations it should support for judges, lawyers, court clerks, administrators, citizens, and system administrators.

**FR1: User Registration and Authentication**

The system shall allow authorized users such as judges, lawyers, court clerks, administrators, and citizens to register and securely log in using their credentials. The system shall verify user identities through authentication mechanisms and provide password recovery and account management features. Role-based access control shall ensure that users can only access functions permitted for their roles.

**FR2: Case Registration and Management**

The system shall enable lawyers and authorized court staff to register new legal cases electronically. Users shall enter case details, upload supporting legal documents, and receive a unique case identification number. The system shall also allow authorized users to update, view, and manage case information throughout its lifecycle.

**FR3: AI-Based Legal Document Classification**

The system shall automatically classify uploaded legal documents using Artificial Intelligence and Natural Language Processing (NLP). Documents shall be categorized according to case type, legal category, or document type, reducing manual effort and improving document organization.

**FR4: Judge Allocation**

The system shall assign cases to judges based on workload, availability, specialization, and court policies. Administrators shall also have the option to manually assign or reassign cases whenever necessary.

**FR5: Intelligent Hearing Scheduling**

The system shall schedule court hearings by considering judge availability, courtroom availability, case priority, and existing schedules. It shall prevent scheduling conflicts and automatically notify all concerned parties about hearing dates and any schedule changes.

**FR6: Case Status Tracking**

The system shall allow lawyers, judges, court staff, and citizens to track the progress of cases in real time. Users shall be able to view the current case status, hearing history, judgments, pending activities, and upcoming hearing schedules.

**FR7: AI-Based Case Duration Prediction**

The system shall analyze historical judicial data and current case characteristics to estimate the expected completion timeline of each case. These predictions shall assist judges and administrators in planning resources and reducing delays.

**FR8: Judicial Workload Analysis**

The system shall monitor and analyze the workload of judges by tracking the number of assigned, pending, and completed cases. Analytical dashboards shall help administrators distribute work more efficiently and identify workload imbalances.

**FR9: Report Generation**

The system shall generate various reports, including case statistics, pending case reports, disposed case reports, judge performance reports, hearing schedules, and workload analysis reports. Reports shall be available in printable and downloadable formats such as PDF.

**FR10: Notification Management**

The system shall automatically send notifications to judges, lawyers, clerks, and citizens through email, SMS, or in-system alerts regarding hearing schedules, case updates, document verification, and judgment announcements.

**FR11: Dashboard Management**

The system shall provide interactive dashboards for different user roles. Dashboards shall display important information such as active cases, pending cases, completed cases, upcoming hearings, judicial workload, and court performance indicators using charts and graphs.

**FR12: User and Role Management**

The system administrator shall manage user accounts, assign roles and permissions, activate or deactivate accounts, and maintain user information. Role-based access shall ensure secure access to system resources.

**FR13: Search and Filter**

The system shall provide advanced search and filtering capabilities, allowing users to search cases using parameters such as Case ID, case type, judge name, lawyer name, filing date, hearing date, and case status.

**FR14: Audit Logging**

The system shall record all important user activities, including logins, case updates, document uploads, hearing scheduling, and administrative actions. Audit logs shall help maintain accountability, transparency, and system security.

**FR15: Backup and Recovery**

The system shall automatically perform periodic database backups and provide recovery mechanisms to restore data in case of system failure, ensuring business continuity and data integrity.

# 5. Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards of the **AI-Powered Smart Court Case Management and Judicial Analytics Platform**. These requirements ensure that the system is secure, reliable, efficient, scalable, and easy to use while meeting the operational needs of the judicial system.

**NFR1: Performance**

The system shall provide fast and efficient performance to ensure smooth judicial operations. It should respond quickly to user requests and support multiple users simultaneously without significant delays.

**Requirements:**

- The system shall respond to user requests within **3 seconds** under normal operating conditions.
- It shall support **10,000 or more concurrent users**.
- Case searches and report generation shall be completed efficiently.
- AI-based document classification should process uploaded documents within a reasonable time.

**NFR2: Security**

The system shall ensure the confidentiality, integrity, and availability of judicial data. Sensitive legal information must be protected from unauthorized access.

**Requirements:**

- Secure user authentication using usernames and strong passwords.
- Role-Based Access Control (RBAC) for different user roles.
- Encryption of sensitive data during storage and transmission.
- Audit logs for all critical activities.
- Protection against unauthorized access and cyber threats.
- Secure HTTPS communication.

**NFR3: Reliability**

The system shall provide consistent and dependable services with minimal failures to ensure uninterrupted court operations.

**Requirements:**

- System availability of at least **99.9%**.
- Automatic backup of case records.
- Recovery from unexpected failures without data loss.
- Consistent operation during heavy workloads.

**NFR4: Availability**

The platform shall be available whenever required by authorized users.

**Requirements:**

- 24×7 accessibility for authorized users.
- Minimal downtime during maintenance.
- Disaster recovery mechanisms to restore services quickly.

**NFR5: Scalability**

The system shall support future expansion as judicial requirements increase.

**Requirements:**

- Ability to support additional courts and users.
- Cloud-based deployment for resource expansion.
- Handle increasing case volumes without reducing performance.

**NFR6: Usability**

The system shall provide a simple, user-friendly, and accessible interface for all users, including judges, lawyers, court staff, and citizens.

**Requirements:**

- Easy-to-understand interface.
- Responsive design for desktops, tablets, and mobile devices.
- Simple navigation with minimal training.
- Accessibility support for differently abled users.

**NFR7: Maintainability**

The system shall be easy to maintain and update throughout its lifecycle.

**Requirements:**

- Modular software architecture.
- Well-documented source code.
- Easy bug fixing and feature enhancement.
- Support for future AI model improvements.

**NFR8: Portability**

The system shall operate across multiple operating systems and web browsers.

**Requirements:**

- Compatible with Windows, Linux, and macOS.
- Accessible through major web browsers such as Chrome, Edge, Firefox, and Safari.

**NFR9: Data Integrity**

The system shall maintain accurate, complete, and consistent judicial records throughout all operations.

**Requirements:**

- Prevent duplicate case entries.
- Validate all user inputs.
- Maintain consistency of legal records.
- Protect data from unauthorized modifications.

**NFR10: Compliance**

The system shall comply with legal and regulatory standards applicable to judicial information systems.

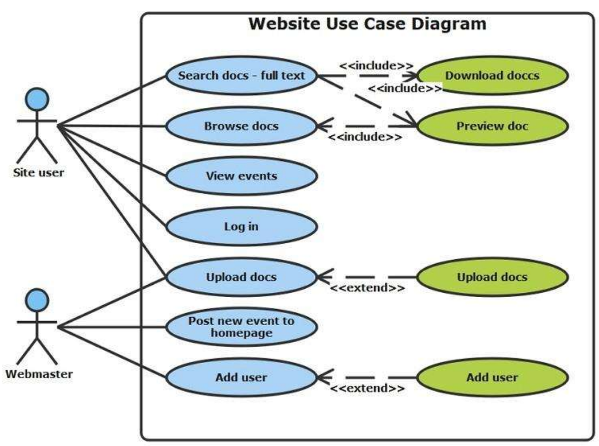
**Requirements:**

- Follow judicial data privacy regulations.
- Maintain confidentiality of legal records.
- Store audit trails for accountability.
- Support compliance with government digital governance policies.

# 6.Actors and Use Cases

## 6.1 Introduction

Actors are the individuals, organizations, or external systems that interact with the **AI-Powered Smart Court Case Management and Judicial Analytics Platform**. Each actor performs specific tasks based on their role and permissions. Use cases describe the various functions or services provided by the system to meet the needs of these actors. Identifying actors and their use cases helps define the system's functional behaviour and ensures that all stakeholder requirements are addressed.



## 6.2 Primary Actors

### 1. Judge

The judge is responsible for handling assigned cases, conducting hearings, reviewing legal documents, delivering judgments, and monitoring case progress. Judges also use analytical dashboards to manage their workload efficiently.

**Use Cases**

- Login to the system
- View assigned cases
- Review legal documents
- Conduct hearings
- Update case status
- Upload judgments
- View AI case duration predictions
- Access judicial analytics dashboard

### 2. Lawyer

Lawyers register new cases, upload legal documents, monitor case progress, and receive hearing notifications through the system.

**Use Cases**

- Register/Login
- File a new case
- Upload legal documents
- View case status
- Receive hearing notifications
- Download case reports

**3. Court Clerk**

Court clerks verify case documents, maintain case records, schedule hearings, and assist judges with administrative activities.

**Use Cases**

- Login
- Verify legal documents
- Update case records
- Schedule hearings
- Modify hearing schedules
- Generate court documents

**4. Court Administrator**

The court administrator oversees the overall functioning of the judicial system by managing judges, monitoring workloads, assigning cases, and generating administrative reports.

**Use Cases**

- Login
- Allocate cases to judges
- Monitor judicial workload
- Generate performance reports
- View analytics dashboard
- Manage court resources

**5. Litigant (Citizen)**

Citizens can access publicly available information regarding their legal cases and receive updates related to hearings and judgments.

**Use Cases**

- Register/Login
- Track case status
- View hearing schedule
- Receive notifications
- Download public documents (if permitted)

**6. System Administrator**

The system administrator is responsible for maintaining the software, managing users, configuring system settings, and ensuring security.

**Use Cases**

- Login
- Create user accounts
- Assign user roles
- Manage permissions
- Backup and restore database
- Monitor system security
- Maintain audit logs

**6.3 Secondary Actors**

**1. AI Engine**

The AI Engine automatically classifies legal documents, predicts case completion timelines, prioritizes cases, and generates analytical insights.

**Use Cases**

- Classify legal documents
- Predict case duration
- Prioritize urgent cases
- Analyze judicial workload

### 2. Notification Service

Responsible for sending automated notifications through email, SMS, or system alerts.

**Use Cases**

- Send hearing reminders
- Notify case status updates
- Send judgment notifications

### 3. Cloud Storage Service

Provides secure storage and retrieval of case records, legal documents, and backups.

**Use Cases**

- Store uploaded documents
- Retrieve case files
- Backup legal records

### 4. Government Judicial Database

Provides integration with government judicial databases for verification and data sharing.

**Use Cases**

- Verify legal records
- Exchange judicial information
- Synchronize official data

# 9. Data Requirements

Data requirements specify the information that the **AI-Powered Smart Court Case Management and Judicial Analytics Platform** must collect, store, process, and manage to support judicial operations efficiently. The system maintains accurate, secure, and organized records of users, legal cases, documents, hearings, judgments, notifications, and analytical reports. Proper data management ensures data integrity, quick retrieval, secure access, and effective decision-making while complying with judicial regulations and privacy standards.

### 9.1 User Data

The system stores information about all registered users, including judges, lawyers, court clerks, administrators, citizens, and system administrators.

**Attributes:**

- User ID
- Full Name
- Username
- Password (Encrypted)
- Email Address
- Phone Number
- User Role
- Address
- Registration Date
- Account Status

**Purpose:**
To authenticate users, manage access permissions, and maintain user profiles.

## 9.2 Case Data

The system stores complete information about every legal case filed in the court.

**Attributes:**

- Case ID
- Case Number
- Case Title
- Case Type
- Filing Date
- Case Category
- Case Status
- Priority Level
- Assigned Judge
- Lawyer Details
- Plaintiff Details
- Defendant Details
- Case Description

**Purpose:**
To manage the complete lifecycle of each court case from registration to closure.

## 9.3 Legal Document Data

The system maintains all legal documents submitted during case registration and court proceedings.

**Attributes:**

- Document ID
- Case ID
- Document Name
- Document Type
- Upload Date
- Uploaded By
- File Format
- Document Category
- AI Classification Result

**Purpose:**
To securely store and organize legal documents while enabling AI-based document classification and retrieval.

## 9.4 Hearing Data

The system records details of every hearing conducted for each case.

**Attributes:**

- Hearing ID
- Case ID
- Hearing Date
- Hearing Time
- Court Hall
- Assigned Judge
- Hearing Status
- Remarks

**Purpose:**
To schedule hearings, avoid conflicts, and maintain hearing history.

## 9.5 Judgment Data

The system stores judgments delivered by judges for completed cases.

**Attributes:**

- Judgment ID
- Case ID
- Judge Name
- Judgment Date
- Judgment Summary
- Final Verdict
- Supporting Documents

**Purpose:**

To maintain official records of court decisions for future reference.

## 9.6 Notification Data

The system stores notifications sent to users regarding case activities.

**Attributes:**

- Notification ID
- User ID
- Notification Type
- Message
- Date and Time
- Delivery Status

**Purpose:**

To maintain records of hearing reminders, case updates, and important announcements.

## 9.7 Judicial Analytics Data

The system stores analytical information generated using AI and historical case records.

**Attributes:**

- Report ID
- Judge Workload
- Pending Cases
- Completed Cases
- Average Case Duration
- Case Trend Analysis
- AI Prediction Results

**Purpose:**

To support judicial decision-making and workload analysis.

## 9.8 Audit Log Data

The system records all important user activities for security and accountability.

**Attributes:**

- Log ID
- User ID
- Activity Performed
- Date and Time
- IP Address
- System Module Accessed

**Purpose:**
To monitor user activities, improve security, and maintain accountability.

## 9.9 Report Data

The system generates and stores administrative and analytical reports.

**Attributes:**

- Report ID
- Report Type
- Generated By
- Generated Date
- Report Format
- Report Summary

**Purpose:**
To provide statistical and performance reports for court administrators.

### 9.10 Data Relationships

The major relationships between data entities are:

- One **User** can manage multiple **Cases** depending on their role.
- One **Case** can contain multiple **Legal Documents**.
- One **Case** can have multiple **Hearings**.
- One **Case** results in one **Judgment** after completion.
- One **User** can receive multiple **Notifications**.
- One **Judge** can handle multiple **Cases**.
- Analytical reports are generated using data from **Cases**, **Hearings**, and **Judgments**.

**Summary of Data Requirements**

| **Data Entity** | **Purpose** |
|---|---|
| User Data | Manage users and authentication |
| Case Data | Store complete case information |
| Legal Document Data | Store and classify legal documents |
| Hearing Data | Schedule and manage hearings |
| Judgment Data | Store final court judgments |
| Notification Data | Send and track user notifications |
| Judicial Analytics Data | Generate AI insights and workload analysis |
| Audit Log Data | Record user activities for security |
| Report Data | Generate court performance and statistical reports |

## 10. External Interface Requirements

External Interface Requirements describe how the AI-Powered Smart Court Case Management and Judicial Analytics Platform interacts with users, hardware devices, software applications, and communication networks. These interfaces ensure smooth communication between the system and external entities while providing a secure, efficient, and user-friendly environment for all stakeholders.

### 10.1 User Interface Requirements

The system shall provide a simple, responsive, and user-friendly web interface that allows different users to perform their respective tasks efficiently. Each user will have a customized dashboard based on their role.

**Features**

- Secure login page for authentication.
- Role-based dashboards for judges, lawyers, clerks, administrators, citizens, and system administrators.
- Online case registration forms.
- Document upload interface.
- Hearing scheduling calendar.
- Case search and filtering options.
- Judicial analytics dashboard with graphs and charts.
- Notification panel for hearing updates and announcements.
- Mobile-responsive design for access through smartphones and tablets.
- Accessible interface supporting users with disabilities.

## 10.2 Hardware Interface Requirements

The system shall operate on standard computing hardware and cloud infrastructure.

**Hardware Components**

- Web/Application Server
- Database Server
- Cloud Storage Server
- Desktop Computers and Laptops
- Mobile Devices
- High-speed Internet Connection
- Backup Storage Devices
- Printers (for official reports and legal documents)

## 10.3 Software Interface Requirements

The platform shall integrate with various software components and third-party services to provide complete judicial functionality.

**Software Interfaces**

- AI Engine for document classification.
- NLP Engine for legal text processing.
- MySQL/PostgreSQL Database Management System.
- Web Server (Apache Tomcat).
- Email Server for notifications.
- SMS Gateway for alerts.
- Cloud Storage Services.
- REST APIs for integration with external judicial systems.
- Government Judicial Database for legal verification.

## 10.4 Communication Interface Requirements

The system shall provide secure communication between users, servers, and integrated services.

**Communication Requirements**

- HTTPS protocol for secure web communication.
- RESTful APIs for external system integration.
- Secure data transmission using SSL/TLS encryption.
- Email notifications.
- SMS alerts.
- Internet connectivity for cloud access.
- Real-time communication between application and database servers.

## 10.5 Security Interface Requirements

To protect sensitive judicial information, the system shall implement secure authentication and authorization mechanisms.

**Security Features**

- Username and password authentication.
- Multi-Factor Authentication (MFA) for administrators (optional).
- Role-Based Access Control (RBAC).
- Data encryption during storage and transmission.
- Automatic session timeout after inactivity.
- Audit logs for all critical system activities.

**10.6 Integration Requirements**

The system shall integrate with external services to improve functionality and interoperability.

**External Integrations**

- Government Judicial Information Systems.
- National Legal Databases.
- Cloud Storage Providers.
- Email Notification Services.
- SMS Gateway Services.
- AI and Machine Learning Services for analytics.

### Summary of External Interface Requirements

| Interface Type | Description |
| --- | --- |
| **User Interface** | Responsive web portal with role-based dashboards |
| **Hardware Interface** | Servers, desktops, mobile devices, printers, backup storage |
| **Software Interface** | AI engine, NLP engine, database, Tomcat, APIs, cloud services |
| **Communication Interface** | HTTPS, REST APIs, email, SMS, SSL/TLS |
| **Security Interface** | Authentication, RBAC, encryption, audit logs |
| **Integration Interface** | Government databases, cloud services, AI services |

# 11. System Features

The **AI-Powered Smart Court Case Management and Judicial Analytics Platform** provides a wide range of intelligent features to improve the efficiency, transparency, and effectiveness of judicial administration. By integrating Artificial Intelligence (AI), Natural Language Processing (NLP), cloud computing, and data analytics, the system simplifies court operations and supports data-driven decision-making.

### 11.1 Online Case Registration

The system allows lawyers and authorized court staff to register new legal cases electronically. Users can enter case details, upload supporting legal documents, and receive a unique Case ID. This eliminates paper-based registration, reduces manual effort, and improves the accuracy of case records.

### 11.2 AI-Based Legal Document Classification

Using Artificial Intelligence and Natural Language Processing (NLP), the system automatically classifies uploaded legal documents based on their content and case type. This reduces manual document sorting, improves search efficiency, and ensures proper organization of legal records.

### 11.3 Intelligent Hearing Scheduling

The platform automatically schedules court hearings by considering judge availability, courtroom resources, and case priority. It minimizes scheduling conflicts, optimizes court resources, and ensures timely hearings for all parties involved.

### 11.4 Real-Time Case Tracking

Authorized users, including judges, lawyers, and litigants, can monitor the status of legal cases throughout their lifecycle. Users can view hearing schedules, pending actions, judgments, and case progress through a centralized dashboard.

### 11.5 AI-Based Case Duration Prediction

The system analyzes historical judicial data and case characteristics to predict the estimated completion time for ongoing cases. These predictions help judges and administrators prioritize cases and reduce delays in court proceedings.

### 11.6 Judicial Analytics Dashboard

The platform provides interactive dashboards displaying key judicial performance indicators such as pending cases, disposed cases, judge workload, hearing schedules, and case trends. Administrators can use these analytics to improve planning and resource allocation.

### 11.7 Automated Notifications

The system automatically sends notifications through email, SMS, or in-system alerts regarding hearing schedules, case status updates, document verification, and judgment announcements. This ensures timely communication with all stakeholders.

### 11.8 Report Generation

The platform generates various reports, including case statistics, pending case reports, judge workload reports, hearing schedules, and performance analysis. Reports can be viewed online or downloaded in printable formats such as PDF.

### 11.9 Role-Based Access Control

Different users are provided with access based on their roles and responsibilities. Judges, lawyers, court clerks, administrators, citizens, and system administrators can only access the information and functionalities authorized for their roles, ensuring data security and confidentiality.

### 11.10 Secure Document Management

The system securely stores all legal documents in cloud-based storage with encryption and backup mechanisms. Authorized users can upload, download, search, and retrieve documents whenever required while maintaining document integrity and confidentiality.

### 11.11 Audit Logging

Every important system activity, such as user login, case registration, document upload, hearing scheduling, and administrative actions, is recorded in audit logs. This improves accountability, security, and compliance with judicial regulations.

### 11.12 Search and Filtering

The platform provides advanced search and filtering options that allow users to quickly locate cases using Case ID, case type, judge name, lawyer name, filing date, hearing date, or case status. This feature significantly improves information retrieval and operational efficiency.

## 12. Assumptions

Assumptions are the conditions considered to be true during the design and development of the **AI-Powered Smart Court Case Management and Judicial Analytics Platform**. These assumptions help define the project scope and expected operating environment.

### A1. Internet Connectivity

It is assumed that all authorized users have access to a stable internet connection to use the web-based platform.

### A2. User Authentication

It is assumed that all users (judges, lawyers, court clerks, administrators, and citizens) possess valid login credentials provided by the judicial authority.

### A3. Availability of Historical Data

It is assumed that sufficient historical court case data is available to train AI models for document classification and case duration prediction.

### A4. Digital Literacy

It is assumed that court staff and other users have basic computer knowledge and can operate the system after minimal training.

#### A5. Cloud Infrastructure

It is assumed that reliable cloud infrastructure is available for secure data storage, backup, and system scalability.

#### A6. Standardized Legal Documents

It is assumed that legal documents are submitted in supported digital formats (such as PDF or DOCX) to enable AI-based processing and classification.

# 12. Constraints

Constraints are the limitations and restrictions that may affect the design, implementation, and operation of the proposed system.

## 12.1 Technical Constraints

- The system depends on a stable internet connection.
- AI prediction accuracy depends on the quality and quantity of historical data.
- Cloud server capacity may limit performance during peak usage.
- Integration with existing judicial systems may require additional APIs and compatibility checks.

## 12.2 Operational Constraints

- Court staff may require training before using the system effectively.
- Existing judicial procedures and workflows must be followed.
- Adoption of the new system may take time due to resistance to change.

## 12.3 Economic Constraints

- Budget limitations may restrict advanced AI features or infrastructure.
- Costs associated with cloud hosting, maintenance, and software licensing must be considered.
- Regular system maintenance and upgrades require financial resources.

## 12.4 Legal Constraints

- The system must comply with judicial regulations and government policies.
- Confidential legal records must be protected according to privacy laws.
- Access to sensitive information must be restricted to authorized users only.
- All digital records must maintain authenticity and legal validity.

## 12.5 Time Constraints

- The project must be completed within the allocated development schedule.
- Testing, deployment, and user training should be completed before system implementation.

# 13. Conclusion

The **AI-Powered Smart Court Case Management and Judicial Analytics Platform** is an innovative LegalTech solution designed to modernize and enhance the efficiency of judicial administration. By integrating **Artificial Intelligence (AI), Natural Language Processing (NLP), cloud computing, and data analytics**, the system addresses the major challenges faced by traditional court management systems, including manual documentation, inefficient hearing scheduling, increasing case backlogs, and limited analytical support.

The proposed platform enables the complete digitization of court case management, from case registration and document handling to hearing scheduling, case tracking, and judgment management. AI-powered features such as legal document classification, case duration prediction, and judicial workload analysis improve decision-making, optimize resource utilization, and reduce delays in court proceedings. Interactive dashboards and analytical reports provide administrators with valuable insights into court performance, enabling better planning and management.

This Software Requirement Specification (SRS) clearly defines the system's objectives, scope, functional and non-functional requirements, actors, use cases, data requirements, external interfaces, assumptions, constraints, and validation. These requirements provide a structured foundation for the successful design, development, testing, deployment, and maintenance of the system while ensuring that stakeholder expectations are fully addressed.

The implementation of this platform is expected to significantly improve the quality of judicial services by increasing transparency, enhancing accessibility, reducing administrative workload, and ensuring secure management of legal records. Citizens will benefit from improved access to case information, while judges, lawyers, and court staff can perform their responsibilities more efficiently through intelligent automation and streamlined workflows.

In conclusion, the **AI-Powered Smart Court Case Management and Judicial Analytics Platform** offers a scalable, secure, and intelligent solution for transforming judicial administration. By leveraging modern technologies, the platform supports faster case processing, better workload management, data-driven decision-making, and improved public trust in the justice system. It represents a significant step toward building a more efficient, transparent, and technology-driven judicial ecosystem capable of meeting future legal and administrative demands.